

Original Article

Machine Learning–Based Prognostic Gene Signature for Early Triple-Negative Breast Cancer

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Purpose This study aimed to develop a machine learning–based approach to identify prognostic gene signatures for early-stage triple-negative breast cancer (TNBC) using next-generation sequencing data from Asian populations.

Materials and Methods We utilized next-generation sequencing data to analyze gene expression profiles and identify potential biomarkers. Our methodology involved integrating various machine learning techniques, including feature selection and model optimization. We employed logistic regression, Kaplan-Meier survival analysis, and receiver operating characteristic (ROC) curves to validate the identified gene signatures.

Results We identified a gene signature significantly associated with relapse in TNBC patients. The predictive model demonstrated robustness and accuracy, with an area under the ROC curve of 0.9087, sensitivity of 0.8750, and specificity of 0.9231. The Kaplan-Meier survival analysis revealed a strong association between the gene signature and patient relapse, further validated by logistic regression analysis.

Conclusion This study presents a novel machine learning-based prognostic tool for TNBC, offering significant implications for early detection and personalized treatment. The identified gene signature provides a promising approach for improving the management of TNBC, contributing to the advancement of precision oncology.

Key words Triple-negative breast cancer, Prognosis, Precision medicine, Machine learning

Introduction

Breast cancer is the most commonly diagnosed cancer among women on a global scale, with the annual estimate for new breast cancer cases worldwide stands at approximately 2.3 million [1]. It is a diverse disease, exhibiting variations in biological behavior, responses to treatment, and prognosis. Triple-negative breast cancer (TNBC) represents a subtype that accounts for 10%-15% of all breast cancer cases and is characterized by the absence of estrogen receptors, progesterone receptors, and human epidermal growth factor receptor 2 expression [2]. Compared to other breast cancer subtypes, TNBC is associated with a poorer prognosis because it lacks the benefit of specific therapies that targets these proteins [3].

Patients with operable stage I-III TNBC undergo a variety of treatment combinations, including neoadjuvant chemotherapy, surgery, adjuvant chemotherapy, immunotherapy, and radiation therapy. Currently, there are no available surrogate biomarkers that can accurately predict prognosis at

the time of diagnosis, allowing for personalized treatment. Although the achievement of pathologic complete response (pCR) is known to provide assistance in predicting future prognosis for patients who have received neoadjuvant chemotherapy [4], the prognostic value of pCR is still limited [5].

The integration of machine-learning and artificial intelligence (AI) into healthcare has led to ongoing research in predicting the prognosis of cancer [6,7]. It is important to utilize bioinformatics techniques for predicting the prognosis of TNBC, given its relatively rare prevalence and the need for long-term follow-up. Previous attempts to predict the prognosis of TNBC using machine-learning algorithms have been made [8-10], but most of them were retrospective studies based on genomic datasets from Western populations, which may not fully capture the clinical characteristics of the Asian patients. Here, we present important genomic features that can predict the prognosis of TNBC by integrating national sequencing data and clinical information.

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Materials and Methods

The model's conceptual diagram is shown in Fig. 1. First, mutations underwent preprocessing and were combined at the gene level. In real-world scenarios, it is challenging to obtain the entire gene set, and not all genes are assumed to be significant. With this in mind, we employed a two-step feature selection process to identify the gene set with the highest predictive power. In the first phase, a random forest model was used to identify the top 10 genes contributing most significantly to prediction. To enhance interpretability and facilitate clinical application, we then applied logistic regression in the second phase to calculate the final risk scores. During this phase, we used p-value-based thresholding, ultimately narrowing down to a final set of five genes (S1 Table).

1. Patients and study design

Among patients who had received treatment for early or metastatic triple-negative breast cancer at Korea University Anam Hospital, those with available next-generation sequencing (NGS) data and clinical response to treatment were chosen for this study. Clinical characteristics such as age, sex, date of diagnosis, molecular subtype, treatment history, and survival data of each patient were collected from their medical records.

2. NGS: KM-00 protocol

The targeted sequencing using tumor tissues was performed using CancerSCAN (Samsung Genome Institute). In patients whose tumor tissues were not available, 10 mL of whole blood was collected with Cell-Free DNA BCT for ctDNA preparation and analyzed by the Axen Cancer Panel (MacroGen). The sequential blood sample was collected within 24 hours before treatment, before the start of the 2nd cycle, time of response evaluation, and at the end of treatment.

Genomic DNA from formalin-fixed paraffin-embedded (FFPE) samples or plasma was extracted using the QIAamp FFPE tissue kit (Qiagen) or QIAamp circulating nucleic acid kit (Qiagen), respectively. Cell-free DNA purity was measured using an Agilent High Sensitivity DNA Kit and a 2100 Bioanalyzer instrument (Agilent Technologies). When required, additional purification was performed using Agencourt AMPure XP (Beckman Coulter) to further remove contaminating nucleic acid. Centrally isolated genomic DNA samples that underwent quality control were sent to the K-MASTER genomic analysis laboratories. K-MASTER used two previously established tissue-based NGS panels (FIRST and CancerSCAN) to detect major genomic aberrations, including mutations, CNAs, and small insertions and deletions in cancer-related genes. CancerSCAN has been further

upgraded to K-MASTER v1.0 and v1.1. Detailed laboratory and bioinformatics protocols are available in the Supplementary Methods.

3. Preprocessing of profile data

The initial genomic datasets underwent preprocessing under the protocol previously described [11]. Following annotation of mutation variants, we applied various criteria to filter potential gene candidates. These criteria consisted of:

- The gene should exhibit an exonic function, falling into categories such as frameshift insertion, frameshift deletion, nonframeshift insertion, nonframeshift deletion, and nonsynonymous single nucleotide variant.
- The ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>) functional description must align with pathologic, likely pathogenic, drug response, risk factor, uncertain significance, or an unknown classification.

Subsequently, the mutations that met these filter criteria were grouped on a gene-level basis. This gene-level categorization was then transformed into a binary classification, where genes with mutations were designated as 1 (positive), while those without mutations were labeled as 0 (negative). This binary representation created a patient-gene mutation variant table. The target label for our analysis pertained to the presence or absence of relapse. Relapse, in this context, was defined as either a recurrence of breast cancer diagnosis or the initiation of chemotherapy subsequent to the initial breast cancer diagnosis. The endpoint of our follow-up period was set at a date in the December 2022. Positive labels were assigned to cases with breast cancer relapse, while negative labels were assigned to patients without any relapse. To facilitate model development and validation, we divided the entire patient-gene mutation variant dataset into a train-test split, with a ratio of 7:3. The split was conducted while taking into consideration the distribution of relapse cases, ensuring a stratified train-test split.

4. Selection of gene candidates and signatures

To begin, we initiated the elimination of highly correlated gene features, employing a threshold of 0.9 for this purpose. Subsequently, for the selection of gene candidates, we undertook the training of a random forest model. Tuning of the hyperparameters for the random forest model was carried out via a 5-fold cross-validation approach. The outcome of this process led to the identification of 10 gene candidates based on the random forest model's analysis.

To compute a risk score, we employed logistic regression using the previously identified set of 10 gene candidates. In this endeavor, we utilized backward elimination steps, implementing a p-value threshold of 0.1 to filter the gene signatures. To facilitate a comparison of the performance

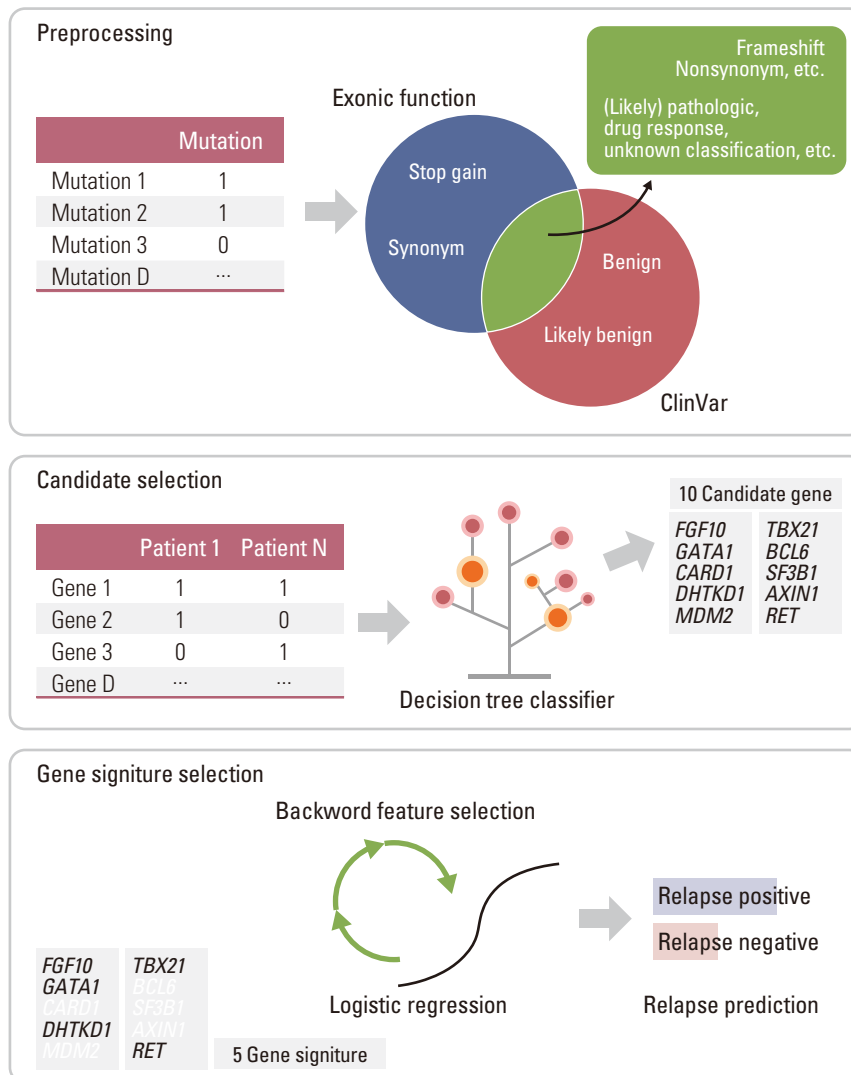


Fig. 1. Gene signature identification for relapse prediction in triple-negative breast cancer (TNBC) patients. An analytical workflow, beginning with variant call format (VCF) files, aggregates gene-level mutations ClinVar and exonic function filtering. A decision tree classifier is employed to identify 10 gene candidates, which are further refined through logistic regression with backward feature selection to ascertain five critical gene signatures. These signatures are determined to predict relapse in TNBC patients with heightened accuracy. Their significance validated by Kaplan-Meier and area under the receiver operating characteristic curves. (Continued to the next page)

between the full model, which utilizes the entire set of gene signatures, and other models that predict relapse by employing individual genes, we fit various logistic regression models with variations corresponding to the utilization of single genes in prediction.

5. Statistical analysis

We evaluated the model's performance by looking at receiver operating characteristics (ROC) curves and the associated area under the ROC curve (AUROC). To understand how well the model predicts survival, we used the Kaplan-

Meier method. We divided patients into two groups based on the median predicted values and then looked at how their survival differed over time. To check if these differences were significant, we used a statistical test called the log-rank test. All of these analyses were carried out using Python 3.9 and R software ver. 4.3.1.

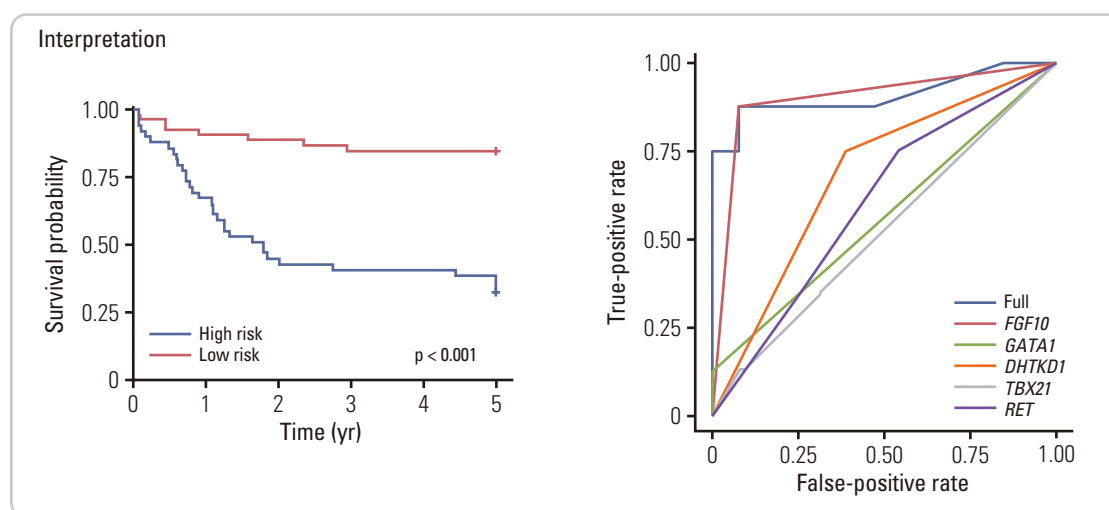


Fig. 1. (Continued from the previous page)

Results

1. Patients' characteristics

A total of 102 TNBC patients were included as either a training for machine-learning or test set for validation of algorithm. The characteristics of the included patients are summarized in Table 1. The median age of the patients was 50 years old. Patients clinical stage III accounted for the highest proportion at 31.4%, while seven patients were initial stage 4. 42.2% underwent surgery after neoadjuvant chemotherapy, 28.4% received adjuvant treatment after surgery, and 7.8% underwent surgery only. NGS was conducted on tissues collected at the initial diagnosis for 68.6% of the patients.

2. Gene candidates and signature selection

The initial dataset contained a total of 892 gene features. After applying a filter to eliminate redundant features due to multicollinearity, we retained 156 gene features. The tuned random forest model demonstrated a classification performance with the following values: AUROC of 0.8269, accuracy of 0.7143, sensitivity of 0.6667, and specificity of 0.7500. From this, we identified 10 gene candidates, specifically *FGF10*, *GATA1*, *CARD1*, *DHTKD1*, *MDM2*, *TBX21*, *BCL6*, *SF3B1*, *AXIN1*, and *RET*. Following a feature selection process known as backward elimination, we ended up with a set of five gene signatures: *FGF10*, *GATA1*, *DHTKD1*, *TBX21*, and *RET*. The details are summarized in Table 2. To calculate the final risk scores, we used the formula:

$$\text{RiskScore} = 1 / \exp(1.3128 \times \text{FGF10} + 2.3964 \times \text{GATA1} + 1.2139 \times \text{DHTKD1} + 2.2145 \times \text{TBX21} + 1.0940 \times \text{RET} - 2.6747).$$

Utilizing the training split dataset, the risk score derived from five gene signatures demonstrated superior predic-

Table 1. Demographic characteristics of study population

Clinical characteristic	No. (%) (n=102)
Age (yr), median (range)	50 (30-85)
Stage	
I	17 (16.7)
II	20 (19.6)
III	32 (31.4)
IV	7 (6.9)
NE	26 (25.5)
Treatment	
Neoadjuvant therapy followed by surgery	43 (42.2)
Surgery followed by adjuvant therapy	29 (28.4)
Surgery only	10 (9.8)
Chemotherapy only	7 (6.9)
NE	13 (12.7)
Tumor status	
Initial diagnosis	70 (68.6)
Surgical tissue after neoadjuvant therapy	26 (25.5)
NE	6 (5.9)

NE, not evaluable.

tive performance compared to other models based on single genes. In Fig. 2A, the prediction performances using the training split set were depicted. The AUROC of the risk score was 0.8327, outperforming individual genes such as *FGF10* (0.7307), *GATA1* (0.6138), *DHTKD1* (0.6571), *TBX21* (0.6138), and *RET* (0.6525). We assessed the classification performance using a test split dataset, presenting the results in Table 3 and Fig. 2B. Our logistic regression model successfully distinguished between relapse and normal cases with AUROC values of 0.9087, as well as an accuracy of

Table 2. Odds ratio of fitted logistic regression

Variable	Odds ratio	Confidence interval		p-value
		Low	High	
<i>FGF10</i>	3.72	1.30	10.93	0.015
<i>GATA1</i>	10.98	1.52	227.98	0.040
<i>DHTKD1</i>	3.37	1.17	10.39	0.028
<i>TBX21</i>	9.16	1.16	199.58	0.067
<i>RET</i>	2.99	0.99	9.76	0.058
Bias	0.07	0.02	0.20	< 0.001

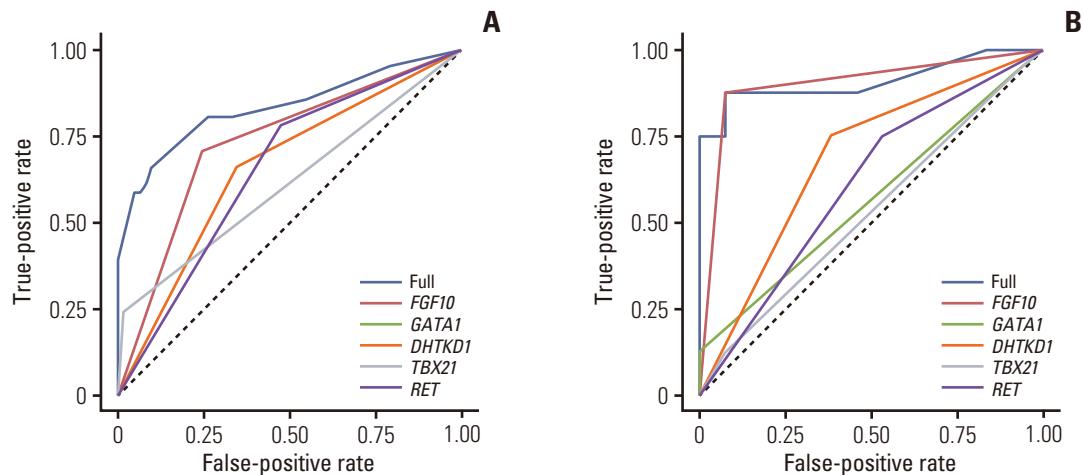


Fig. 2. Area under the receiver operating characteristic (AUROC) curves of predictive models. Panel A illustrates the AUROC curve derived from the training dataset, while Panel B from the test dataset, each corresponding to the 5-gene signature-based logistic regression model (full model) and individual gene models for *FGF10*, *GATA1*, *DHTKD1*, *TBX21*, and *RET*. Panel A indicates the full model's superior performance over single gene-based models, whereas Panel B exhibits the *FGF10* single gene-based model's performance, comparable with the full model.

0.9048 (sensitivity, 0.8750; specificity, 0.9231; positive predictive value [PPV], 0.8750; negative predictive value [NPV], 0.9231). Notably, when considering the performance of a model that relied solely on *FGF10*, it exhibited a remarkable performance (AUROC, 0.8990; accuracy, 0.9048; sensitivity, 0.9750; specificity, 0.9231; PPV, 0.875; NPV, 0.9231). However, the other gene signatures showed lower performance when used individually to predict relapse, with AUROC values below 0.7 for *GATA1* (AUROC, 0.5625; accuracy, 0.6667; sensitivity, 0.1250; specificity, 1.0000; PPV, 1.0000; NPV, 0.6500), *DHTKD1* (AUROC, 0.6827; accuracy, 0.6667; sensitivity, 0.7500; specificity, 0.6154; PPV, 0.5455; NPV, 0.8000), *TBX21* (AUROC, 0.5240; accuracy, 0.6190; sensitivity, 0.1250; specificity, 0.9231; PPV, 0.5000; NPV, 0.6316), and *RET* (AUROC, 0.6058; accuracy, 0.5714; sensitivity, 0.7500; specificity, 0.4615; PPV, 0.4615; NPV, 0.7500).

In Fig. 3, we visually represented the expression of the five gene signatures using a heatmap. This visualization helped

distinguish expression patterns between normal and relapse cases in TNBC patients. It's observable that five gene signatures exhibit a comparatively higher likelihood of mutation within the recurrence group.

3. Survival analysis of the prognostic gene signature in relapse of TNBC

To delve deeper into the connection between gene signatures and the likelihood of relapse in TNBC patients, we conducted a survival analysis using the Kaplan-Meier statistical method. This analysis revealed that the entire set of gene signatures exhibited a noteworthy association with the relapse of TNBC patients, as indicated by p-values below 0.01 (refer to Fig. 4B-F). Furthermore, the results from our logistic regression analysis indicated a substantial distinction in relapse risk between patients categorized as high and low risk (using a threshold of 0.45), with a p-value below 0.01 (Fig. 4).

Table 3. Classification performance on the test dataset

Model	Threshold	AUROC	Accuracy	Sensitivity	Specificity	PPV	NPV
Full	0.45	0.9087	0.9048	0.8750	0.9231	0.8750	0.9231
<i>FGF10</i>	0.5	0.8990	0.9048	0.8750	0.9231	0.8750	0.9231
<i>GATA1</i>	0.5	0.5625	0.6667	0.1250	1.0000	1.0000	0.6500
<i>DHTKD1</i>	0.5	0.6827	0.6667	0.7500	0.6154	0.5455	0.8000
<i>TBX21</i>	0.5	0.5240	0.6190	0.1250	0.9231	0.5000	0.6316
<i>RET</i>	0.5	0.6058	0.5714	0.7500	0.4615	0.4615	0.7500

AUROC, area under the receiver operating characteristics; NPV, negative predictive value; PPV, positive predictive value.

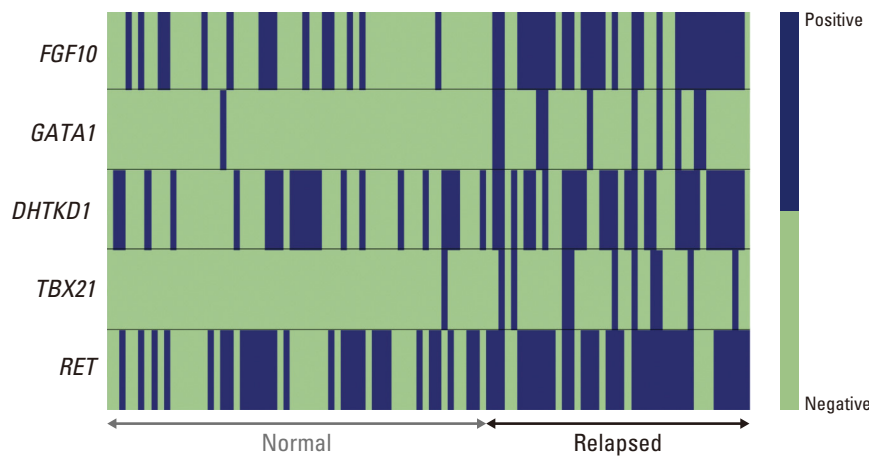


Fig. 3. Heatmap depicting expression of the five gene signatures. This heatmap contrasts the expression levels of five gene signatures between patient groups categorized by relapse and normalcy.

Discussion

With the two-stage gene signatures identifications, we could select five most significant genes to predict relapse in TNBC patients with considerable prediction performances (AUROC, 0.8990). The gene signatures were *FGF10*, *GARA1*, *DHTKD1*, *TBX21*, and *RET*. With the following survival analysis, each gene signatures showed diverging patterns (Fig. 4).

First of all, the *FGF10* gene encodes a protein called fibroblast growth factor 10 (*FGF10*), which plays crucial roles via *FGFR2b* signaling in both the epithelium and mesenchyme during mammary gland development [12]. In humans, *FGF10* expression has been observed in 10% of breast adenocarcinomas, and the activation of *FGFR2b* signaling is associated with a poorer prognosis [12]. Besides breast cancer, *FGF10* also known to stimulate cell migration and invasion in pancreatic cancer cells through its interaction with *FGFR2-IIIb*, resulting in aggressive tumor proliferation [13].

GATA-binding factor 1, or GATA-1, is the founding member of the GATA family of transcription factors and is also referred to as the erythroid transcription factor. GATA factors

are zinc finger DNA-binding proteins that regulate the development of various tissues by either activating or repressing transcription [14]. Consequently, GATA factors play a crucial role in coordinating cellular maturation, proliferation arrest, and cell survival [15]. They have been associated with colorectal and ovarian cancer progression, as well as gemcitabine resistance in pancreatic cancer [16-18]. In breast cancer, there have been reports suggesting that GATA-1 induction may lead to epithelial-mesenchymal transition, which is considered an indicator of poor prognosis [19].

The *DHTKD1* gene encodes a component of a mitochondrial 2-oxoglutarate-dehydrogenase-complex-like protein involved in the degradation pathways of several amino acids, including lysine [20]. Mutations in this gene are associated with 2-aminoadipic 2-oxoadipic aciduria and Charcot-Marie-Tooth Disease [20]. Research related to cancer progression, particularly in breast cancer prognosis prediction, is rare. However, in a recent study on bladder cancer, it was found that circDHTKD1 was positively associated with lymph node metastasis and significantly upregulated in bladder cancer [21]. *DHTKD1* deficiency leads to mito-

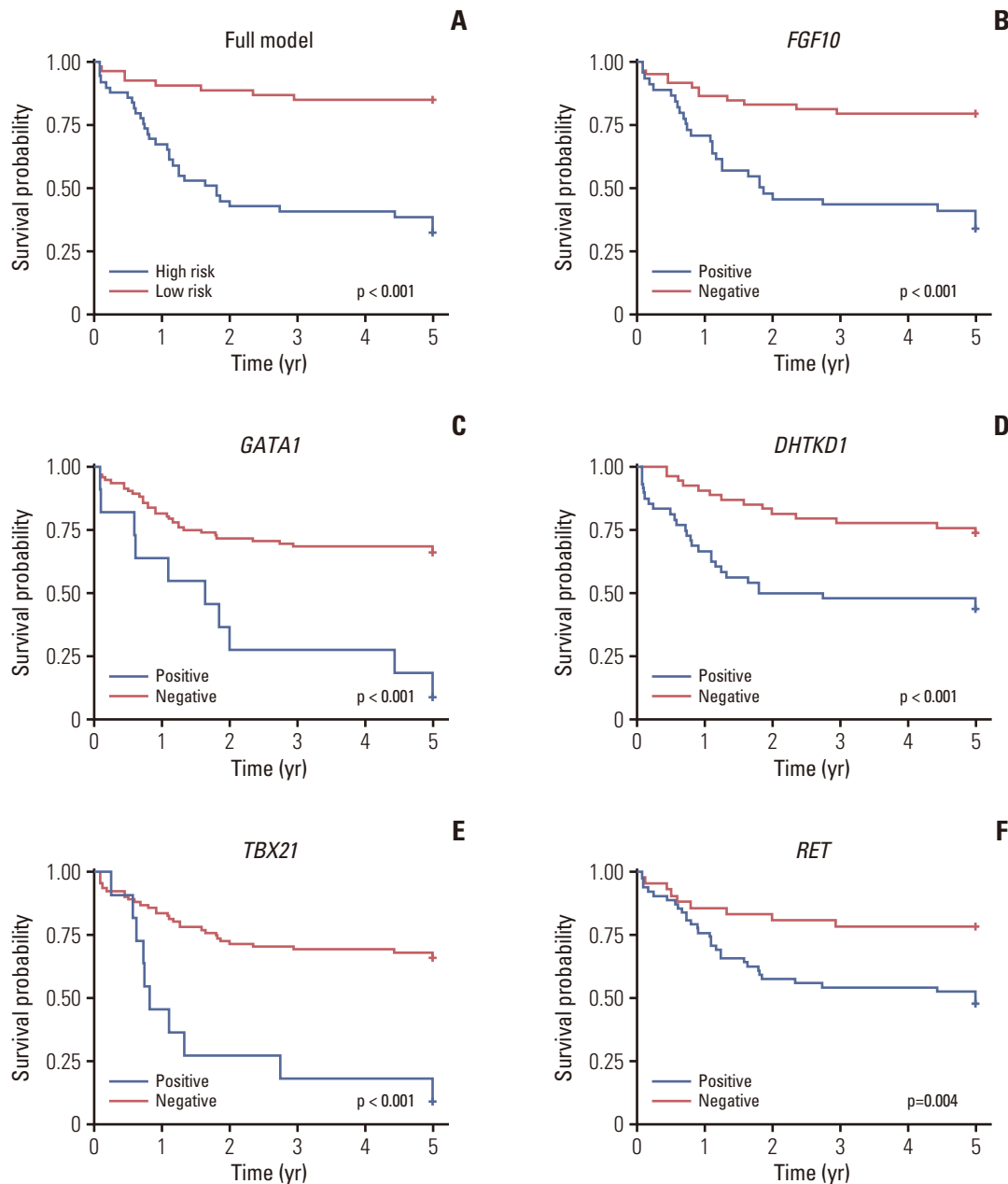


Fig. 4. Kaplan-Meier survival analysis for gene-based models. (A) The survival probabilities using the full model with five gene signatures. (B-F) The prognostic estimations based on single gene models for *FGF10*, *GATA1*, *DHTKD1*, *TBX21*, and *RET*. Each survival curve distinctly demonstrates the divergence between the positive and negative (high risk and low risk) groups.

chondrial dysfunction [22], which in turn can contribute to cancer progression by altering mitochondrial metabolism [23]. This alteration can lead to an increase in the production of mitochondrial reactive oxygen species and changes in cellular redox status. This, in turn, affects the activities of transcription factors such as HIF1 α and FOS-JUN, leading to changes in gene expression and the stimulation of cancer cell proliferation [24].

TBX21 gene belongs to a phylogenetically conserved gene family characterized by a common DNA-binding domain known as the T-box [25]. Studies conducted in mice have demonstrated that the Tbx21 protein serves as a Th1 cell-specific transcription factor responsible for regulating the expression of the hallmark Th1 cytokine, interferon- γ [26]. Recently, Zhao et al. [27] suggest that *TBX21* may be used to assess the survival of patients and promote cancer stemness

through the TBX21–IL-4 pathway in lung adenocarcinoma. Further investigation is needed to elucidate the additional roles of immunotherapy in TNBC, where the response to immunotherapies is becoming a key prognostic factor.

The *RET* gene, located on human chromosome 10q11.2, encodes a transmembrane receptor and belongs to the tyrosine protein kinase family of proteins [28]. *RET* gene alterations, including amplification, missense mutations, known fusions, novel fusions, and rearrangements, have been observed in breast cancer, confirming its involvement in the development of breast cancer [29]. Previous studies have shown that the receptor tyrosine kinase *RET* is overexpressed in a subset of estrogen receptor (ER)–positive breast cancers, and the interaction between *RET* and ER plays a crucial role in responses to endocrine therapy [30]. *RET* expression varies significantly among different breast cancer subtypes, with the lowest *RET* expression in the basal-like subtype and the highest expression in the luminal A subtype [31]. Further research is needed to understand the role of *RET* in TNBC, particularly in relation to its molecular subtypes.

Attempts to predict the prognosis of TNBC have been made by classifying its subtypes. Lehmann classified TNBC into six subtypes based on gene expression profiles [32] and later refined this to four: basal-like 1 (BL1), basal-like 2 (BL2), mesenchymal (M), and luminal androgen receptor subtype [33]. Multi-omics analysis was also proposed to suggest the immune cell composition of each subtype and differential pharmacological vulnerabilities [34]. Although various classification methods have been proposed to address the heterogeneity of TNBC, they have not yet achieved perfect precision treatment [35,36].

This clinical unmet need led to attempts to incorporate AI [37]. Alsaleem et al. [38] employed artificial neural network techniques to predict distant metastasis-free survival and breast cancer-specific survival in TNBC patients, discovering a two-gene prognostic signature (*ACSM4* and *SPDYC*). This signature showed good predictive power with a probability of survival hazard ratio of 0.60 (95% confidence interval, 0.47 to 0.76) when high compared to low. However, the identified genes *ACSM4* and *SPDYC*, which are involved in CDK1/2 expression and the metabolism of carboxylic acids and fatty acid beta oxidation, respectively, did not show a clear association with TNBC biology.

In the process of identifying a gene set that effectively explains the prognosis of TNBC and its cancer biology, we selectively curated a series of mutation groups. One such decision involved excluding nonsense (stopgain) mutations. In the analysis that included all stopgain mutations, the derived top 5 gene set predominantly consisted of oncogenes, making it difficult to explain cancer biology through a loss-of-function perspective. This inclusion also tended to

diminish the overall predictive accuracy (AUROC, 0.8269 to 0.8173). Based on prior knowledge of each gene's functional relevance, we concluded that the original gene set (*FGF10*, *GATA1*, *CARD1*, *DHTKD1*, *MDM2*, *TBX21*, *BCL6*, *SF3B1*, *AXIN1*, and *RET*) provides greater insight and interpretability.

This study has several limitations. Our focus was solely on gene-level mutations categorized as binary labels, employed for analytical purposes. However, this approach may limit the depth of information captured by the model. There exists an opportunity for enhancement by integrating diverse data types, such as clinical data and digital pathology images. Thus, our future research endeavors aim to develop a comprehensive model capable of accommodating both gene signatures and the varied data modalities present in this study. Another point is the relatively small dataset used in this study, comprising only 102 samples, which resulted in a limited test set. While the full model slightly outperformed in terms of AUROC, the *FGF10*-only model also demonstrated nearly comparable performance. This may indicate the strong predictive power of *FGF10*; however, due to the limited size of the dataset, this performance may not be fully validated. Consequently, we are planning future studies to further validate this model in a larger data collection environment or with external datasets.

To our knowledge, this is the first study to use machine learning to analyze a cohort exclusively comprising Asian TNBC patients. It also boasts the advantage of using targeted NGS, which is actively utilized in clinics. Our model exhibited noteworthy predictive performance despite employing only five genes with minimal sequencing and computational expenses, contrasting with previous studies that used complex methods with long turnaround times. Three of the five genes we identified (*FGF10*, *GATA-1*, *RET*) have already been confirmed to have direct or indirect influences on cancer biology. Additionally, *TBX21* is associated with an immune signature, lending credibility to its role in predicting TNBC prognosis. Consequently, this approach holds promise for practical implementation within real-world clinical settings.

Electronic Supplementary Material

Supplementary materials are available at Cancer Research and Treatment website (<https://www.e-crt.org>).

Ethical Statement

The research protocol was approved by Institutional review board of Korea University Anam Hospital (IRB No. 2023AN0080). Informed consent was obtained from study participants for genomic analysis and the presentation of research finding. This study was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines.

Author Contributions

Conceived and designed the analysis: Park KH.
 Collected the data: Kim JW, Park KH.
 Contributed data or analysis tools: Ahn S.
 Performed the analysis: Lee J, Lee SH, Ahn S.
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Conflicts of Interest

Conflict of interest relevant to this article was not reported.

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